



Financial Institutions & Markets- ENT 309

PPT #6, IITD, Summer, 2026



Revisiting Imp Terms

- Interest Rate: How many?
 - Simple, Compounding, Reducing Balance
 - Short-term, long-term interest rates in the economy
 - Fixed rate, Floating rate
 - Layman Language: What is earned or paid above principal/loan
- YTM Yield to Maturity:
 - Layman Language: Rate of Earning from varying cash flows.
 - Rate at which each 'smaller packets' of Principal amount are compounded (refer earlier discussion related to bonds).
 - Discount rate at which NPV of CFs = Market Price.

Discount Rate/ Bond Valuation



- Example: Face value = ₹100, Coupon = 8%, Maturity = 5 years, Market price = ₹102. Will you buy it?
- A fair required return expected is 7%. Opportunity Cost. Discount Rate?
- Valuation:

$$P = \sum_{t=1}^5 \frac{8}{(1.07)^t} + \frac{100}{(1.07)^5}$$

- P (**Bond Value** at a discount rate = Fair required return) = INR 104
- Market P of 102 is < **Bond Value** of 104. Hence Undervalued. May be a good buy.

Discount Rate/ Bond Valuation



Comparing 2 Bonds

- Should you buy a XYZ Finance NCD at a price of INR X or buy a Gol 10-year bond at a price of Y? Face Value & Coupon Rate is given.
- Will you calculate & compare YTM & Choose the bond with higher YTM.

- Yes/No. Why?

Bond	Yield
AAA company	7.5%
Distressed company	14%

- How do you take the risk (default + liquidity) into account?
- Choose a discount rate = risk free rate + risk premium/spread (based on the risk rating of the bond).

Discount Rate/ Bond Valuation



- Comparing two bonds
 - Calculate 'Bond Value' based on a '**chosen discount rate**'.
 - If 'Bond Value' is $>$ Market Price of bond, it is Undervalued
 - If 'Bond Value' is $<$ Market Price of bond, it is Overvalued
- Any Other Factors to be considered??
- Yes. Sensitivity to change in Interest Rates. Depends on ?
- Compare Bonds with similar
 - Maturity
 - Credit Rating
 - Liquidity
 - Other characteristics (Senior/Junior, Secured/Unsecured etc.)

Discount Rate/ Bond Valuation

Example: Market Yield for Different
Rated Bonds

Bond Rating	Market Yield
Government	6.8%
AAA	7.6%
AA	8.3%
A	9.2%

*The discount rates will also follow a trend similar to
Market Yield- Low for Govt and higher as we move to
riskier bonds with lower credit ratings*

Slide included at the end of PPT 5 also

Credit Rating Scales by Agency, Long-Term

Moody's	S&P	Fitch	
Aaa	AAA	AAA	Prime
Aa1	AA+	AA+	High grade
Aa2	AA	AA	
Aa3	AA-	AA-	
A1	A+	A+	Upper medium grade
A2	A	A	
A3	A-	A-	
Baa1	BBB+	BBB+	Lower medium grade
Baa2	BBB	BBB	
Baa3	BBB-	BBB-	
Ba1	BB+	BB+	Non-investment grade speculative
Ba2	BB	BB	
Ba3	BB-	BB-	
B1	B+	B+	Highly speculative
B2	B	B	
B3	B-	B-	
Caa1	CCC+	CCC	Substantial risk
Caa2	CCC		Extremely speculative
Caa3	CCC-		Default imminent with little prospect for recovery
Ca	CC	CC	
	C	C	In default
C			
/	D	D	
/			

"Junk"



Agenda

1

FOREIGN EXCHANGE MARKET

- A bit of History
- Foreign Exchange Market: Overview
- Forex Market: Operations
- Forex Market Risks
- Determining Currency Exchange Rates



A Bit of History

- Historically coins and notes issued by kings had some precious metal like gold & silver. The amount and value of the metal determined the value of a particular currency.
- Gold Coins issues by Chandragupta 1 of Gupta Dynasty





A Bit of History

- This practice carried over to the modern era.
- Under the Gold Standard, countries made their currency convertible on demand to gold at a fixed conversion rate.
- Most major economies UK, France, Germany, USA etc.
- Problems with Gold Standard
 - Volume of currency in the economy linked to the amount of gold with the country
 - Put a constraint on rapid growth. If money supply does not keep pace with growth in the economy, the prices will fall. Why?
- World War 1 (1914-1919) – Gold Standard suspended. Lots of money needed to be printed to finance war efforts



A Bit of History

Post World War 1:

- Massive inflation spike in the US, Germany, UK etc.
- Followed by slowing down of economic growth/mild depression
- Countries resistant to go back to Gold Standard. Unstable.
- US hit by the Great Depression in 1931
- General turmoil in Europe. Economic struggle, Political changes, social transformation
- World War 2: 1939-1945



A Bit of History

Post World War 2:

- US became the most powerful country in the world.
- Bretton Woods System (1944-1971):
 - The US dollar was fixed at \$35 per ounce of gold.
 - Other currencies were fixed to the US dollar.
 - Only foreign central banks could convert dollars into gold
 - All countries agreed to the above.
 - World Bank & IMF established
 - UN established.
- USD became the 'MAIN CURRENCY OF THE WORLD'.



A Bit of History

- US- Vietnam War: 1955 to 1973.
- US printed a lot of dollars and spent them.
- Countries which had stock of dollars with them worried about its value. Demanded Gold in exchange.
- US did not have the equivalent amount of gold.
- 1971: US President Richard Nixon suspended Bretton Woods. No more linkage of USD to Gold.
- Every country's currency became FIAT currency.
- Value of a fiat currency comes from the strength of the economy and market confidence.

2

Foreign Exchange Market



Foreign Exch Market

- Why is FX needed?
 - For international trade of goods and services
 - FX is a financial asset- Companies and people invest in FX to earn returns
 - FX investment volumes are higher than FX for trade volumes
 - FX market is the LARGEST financial market in the world. Bigger than global equity market.
- Which are the top 10 currencies for FX trade?
 - USD, EUR, JPY, GBP, CNH, AUD, CAD, CHF, HKD, NZD
- Top 3 pairs: EURUSD, USDJPY, GBPUSD



Foreign Exch Market

- More than 50 currencies traded regularly.
- USD leads by far. Nearly 90% of transactions have one side in USD.
- Attempts being made to move away from USD. Examples:
 - INR/ Ruble Trade
 - Trading among BRICS countries
 - 11 full member countries. Brazil, Russia, India, China, South Africa, Egypt, Ethiopia, Iran, Saudi Arabia, the UAE, and Indonesia PLUS 10 associate members.
- Difficult to shift from USD. Why?

Trump repeats tariffs threat to dissuade BRICS nations from replacing US dollar



January 31, 2025

- Trump warns BRICS against replacing U.S. dollar with tariffs threat
- BRICS discussions on new currency gain momentum post-Russia sanctions
- U.S. dollar remains primary reserve currency, study shows

Foreign Exchange Market: Operations



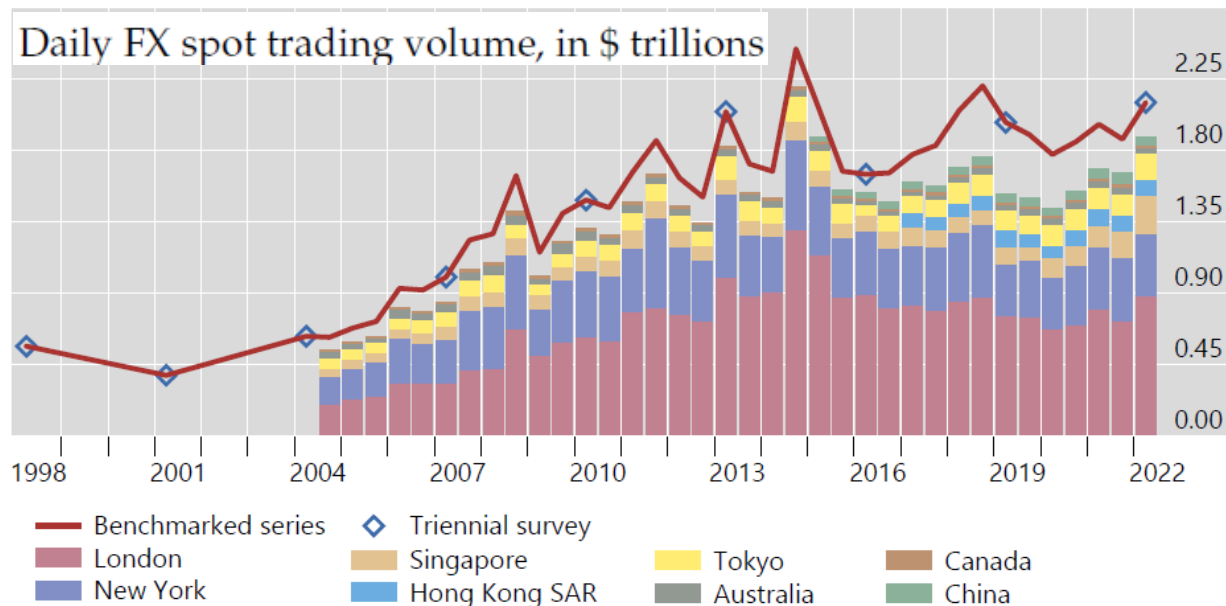
FX Market: Operations

- OTC Market:
 - FX market is an over-the-counter market
 - No central exchange or authority
 - Compare to how stocks and bonds are traded
- Why is it OTC?
 - 24 hours market. Transactions happen across the world.
 - No single jurisdiction
- FX spot trading concentrated in a few bigger financial centres
 - London, NY, Singapore, HK. Why?
 - FX as investments asset is far a larger market than FX for trade
 - Data centers/matching engines hosted for e trading. London, NY, Tokyo, Singapore



FX Market: Operations

- Spot Transaction: The currencies are exchanged immediately.
- Forward Transactions: The currencies are exchanged at a future date

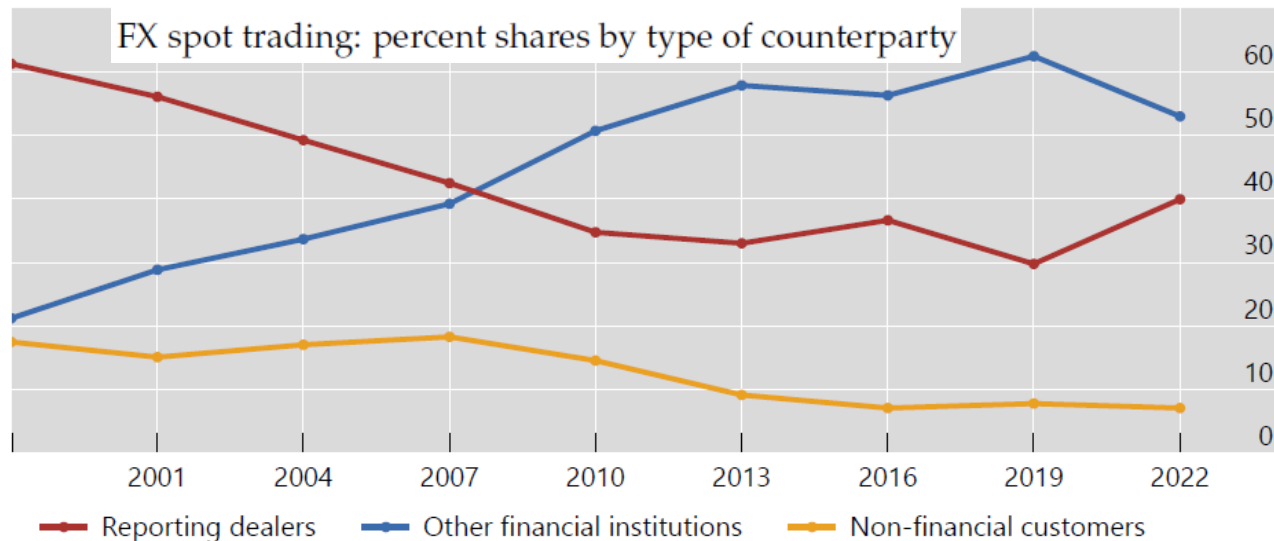


FX Market: Operations



- Two distinct segments: FX dealer to customer & Interdealer
- Main counter-parties of FX Bank-Dealer: Other Bank- Dealers, Financial Institutions (institutional investors, asset managers, hedge funds, commodity trading advisors, HFTs, central banks), Non-Financial (corporates using FX for trade)

FX spot trading: percent shares by type of counterparty

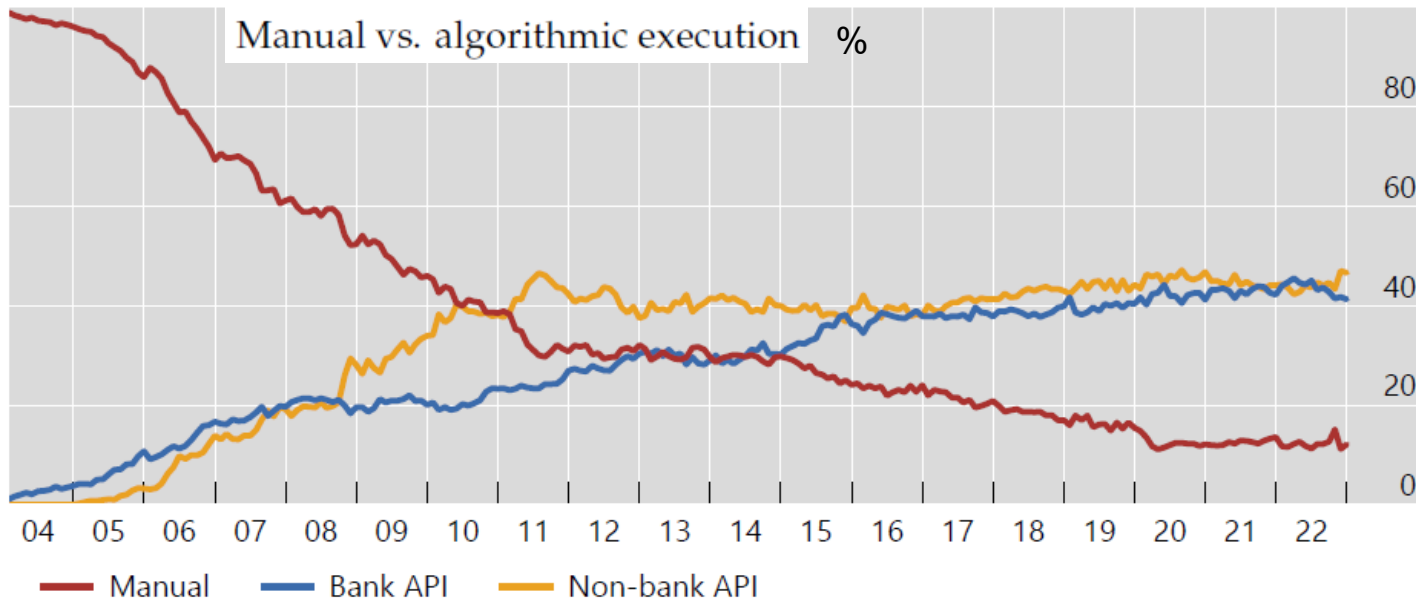


- Share of Non-Financial down from 20% to 10%
- Share of Financial Inst is up to >50%
- FX trading driven by Financial Institutions

FX Market: Operations



- Platforms which give quote from multiple dealers have become popular
- Algorithmic Trading has grown tremendously
- Two types of Algos: Execution Only or Decision Making too.



FX Market: Operations



- An Indian importer of Soybean wants to pay a Brazilian exporter.
 - The importer's bank may not have a branch in Brazil.
 - May not maintain BRL (Brazilian Real) accounts.
 - May not have a direct relationship with the Brazilian bank.
- How will the transaction happen?

FX Market: Operations



- Importer instructs Bank A (say SBI) in India to make payment.
- Bank A converts INR into a major international currency, often USD.
- Bank A sends USD through a correspondent bank (say Citibank) in the United States.
- Citibank might have relations with a Brazilian bank or with another bank that has relations with the Brazilian bank. Transfers USD.
- The Brazilian bank converts USD into BRL. Exporter receives BRL.

Importer → Indian Bank → US Correspondent Bank → Brazilian Correspondent Bank → Brazilian Bank → Exporter

FX Market: Operations



- Correspondent Bank: Holds deposits on behalf of another bank and performs services for it.
 - SBI maintains a USD account with Citibank in USA
 - Citibank is SBI's correspondent bank
- SBI's account with Citibank USA is a **Nostro account** (our account with you)
- From Citibank USA's perspective, SBI account with them is a **Vostro account** (Your account with us)
- There can be multiple correspondent banks in the transaction to set up the pipe for FX flow.

FX Market: Operations



- How are the rates quoted? $\text{USD/INR} = ?$ $\text{INR/USD} = ?$
- 1 USD = How many INR & 1 INR = How many USD
- Dealers- Banks quote Bid/Ask rates
- Bid Rates: Dealer ready to Buy foreign currency
- Ask or Offer Rates: Dealer ready to Sell foreign currency
- Difference in Bid/Ask rates is the 'spread' made by the dealer.
- Bid rate: USD/INR 94.50, Ask Rate: USD/INR 94.55
- Spread: INR 0.05

FX Market: Operations



- Bid is always lower than Ask/ Offer. Why?
- Dealer buys FX and then sells it. To make a profit, he has to buy at a lower price and sell at a higher price.
- Dealer buys at Bid price & sells at Ask price. Hence $\text{Bid} > \text{Ask}$.
- What does 'Spread' cover?
- It covers different risks in the FX process and the profit of the dealer.

Forex Market Operations: Credit/Counterparty/Settle ment Risk

Forex Operations: Risk



Credit Risk / Counterparty Risk/ Settlement Risk

- Two parties to a transaction. Party & Counterparty.
- Person who initiates the transaction (Buyer or Seller) is the party
- Other side (seller or buyer as the case may be) is the counter-party
- How are FX Spot transactions settled between dealers-banks? Do currencies change hands?
- No, deposits in currencies with the banks are used to settle accounts
- Large sums of money involved. Settlement at T+2 (transaction+2) days.
- So, credit extended for 2 days. Leads to credit/ counterparty risk



Forex Operations: Risks

Credit Risk / Counterparty Risk/ Settlement Risk

- **Herstatt Risk:** 26th June, 1974 **Bankhaus Herstatt**, a German bank was shut down by regulatory authorities due to some reasons.
- It had received payment of Deutschemarks but had not paid out USD
- The global FX froze. Erosion of trust. No one willing to take counterparty risk.
- It became a byword for counterparty risk

Forex Operations: Risks

Solution #1: To introduce intermediaries to transfer risk

- **Prime brokerage:** Started becoming popular in 1990s. Run by large, global banks.
 - Clients: Small banks, asset managers, foundations, pension funds etc.
 - Offer Anonymity: Trades done under the name of the Prime Broker
 - Offer Leverage: For a deposit of 100, the client can trade X times of 100. So, provide credit/ loan at a fee for trading larger volumes of FX.
 - Credit Risk: Is of the two banks who are Prime Brokers rather than of the clients of the two banks who are involved in the transaction
 - Better Price Discovery: The clients can deal with more dealer-banks under the umbrella of their Prime Broker

Forex Operations: Risks

Solution #2: New INTERMEDIARY for simultaneous funds transfer

- **Intermediary**: To bring some features of an exchange
- G-10 central banks launched Continuous Linked Settlement (CLS) in 2002
- CLS is a Specialized Institution. Regulated by Federal Reserve Board, USA
- CLS settles FX transactions on a Payment versus Payment (PvP) Basis - payment in both currencies happens simultaneously.
- Takes care of the settlement risk
- India is not a direct member of CLS. Indian Domestic banks access CLS through The Clearing Corporation of India (CCIL)

Determining Currency Exchange Rates



Importance of FX rate

- Change in FX rate changes the costs in the country relative to the other country
- Say USD/INR moved from 90 to 95.
- This is **Depreciation of INR**- its value has gone down
- For Importer: Good or Bad?
- For Exporter: Good or Bad?
- For a normal consumer: Good or Bad?
- How will these 3 be impacted if INR **appreciates**?



Exchange Rates

THEORY: LAW OF ONE PRICE

- Price of an **identical good** should be same all over the world IF transportation costs very low and no other barriers like tariffs.
- Example: Price of a pen in India is INR 500 when USD/INR is 100. So, price of the same pen in the USA would be \$5.
- The price in India become INR 550. Price unchanged in USA
- What is New Exchange Rate?
- $\text{INR } 550 \sim \$5$. Hence USD/INR now is INR 110.
- 10% increase in price in INR translates to 10% depreciation in USD/INR



Theory of PPP

THEORY: PURCHASING POWER PARITY

- A prominent theory of how Exchange Rates are determined
- **Exchange Rate** between 2 countries changes to reflect the change in the Price Levels of the two countries.
- Law of One Price applied to Country Price Levels.
- Review example of change in price of pen.
- If the price level of Country A rises relative to Country B, the currency of Country A will depreciate.
- Above example: Country A: India, Country B: USA



Theory of PPP: RER

Real Exchange Rate (RER)

- Exch Rate is normally quoted in NOMINAL EXCHANGE RATE terms
- REAL EXCHANGE RATE takes into account the price levels in the two countries.
- Somewhat like Real and Nominal interest rates
- Example: Cost of 1 Pepsi can in India is INR 500. Nominal USD/INR = 100. So, it is \$5. Let us say cost of Pepsi Can in USA is USD 6.
- Real Exchange Rate (RER) = $5/6 = 0.83$
- Unit of RER ?



Under/Over Valued Currency

- What should happen when RER is 0.83?

A THOUGHT EXPERIMENT

- Pepsi is cheaper in India. But it & sell in USA. Make \$1 profit,
- Sell \$ to buy INR to buy Pepsi in India and export to USA.
- Exports from India grow, Supply of \$ increases & Demand for INR goes up.
- INR becomes expensive, USD become cheaper. USD/INR appreciates.
- **Conclusion:** Whenever $RER < 1$, that currency is UNDERVALUED
- If currency is correctly valued, then $RER = 1$.
- If $RER > 1$, currency is OVERVALUED.

Big Mac Index



- BMI invented by Economist is 1986. Compares the price of Big Mac Cheese Burger in different countries.
- BMI Extrapolates to – Is the currency at the 'correct level'?
- <https://worldpopulationreview.com/country-rankings/big-mac-index-by-country>
- How many Big Macs can you buy in country A compared to USA for the same amount of USDs?

The Soros Saga



- Sep 16, 1992
- Is the exchange rate 'correct' ?
- <https://youtu.be/QibhEjxLT1U?si=dK0OF4BsXJEEUWU->



PPP Partially explains FX Rate



Three Main Reasons Why PPP Partially Explains FX Rates

- Price rise of non-tradeable goods (goods like houses and services like JEE coaching, taxi service etc,) which are not exported will not impact exchange rate.
- Differentiation in goods. Honda City, Baleno models sold in India are different than their global counterparts. So, change in the price of Baleno in Japan will not impact Yen/INR exchange rate.
- Assumption of no tariffs or quota not valid. Tariff is a tax applied in imports. Quota is a limit on the quantity allowed to be imported.
 - Both raise the domestic price of imports in the importing country
 - Distort the market.



UNION BUDGET 2025-26 PROPOSES TO REMOVE SEVEN CUSTOMS TARIFF RATES FOR INDUSTRIAL GOODS

U.S. Tariff Actions Since January 20, 2025: Timeline and Status

Table 2. U.S. Tariff Actions Since Jan. 20, 2025: IEEPA

As of December 31, 2025

Fentanyl-related action on goods from Canada (Executive Order [\[E.O.\] 14193](#), as amended)

Countries Affected	Canada
Products Affected	All goods <i>Exception:</i> imports that claim and qualify for duty-free preference under the U.S.-Mexico-Canada Agreement (USMCA).
Current Tariff Rates	35% for Canadian goods that do not claim or qualify for USMCA duty-free preference; 10% for energy and potash imports that do not claim or qualify for USMCA duty-free preference; 40% for goods "transshipped to evade applicable duties."
Current Status	35% on most goods and 40% on transshipped goods, effective as of August 1, 2025. (Initial rate (of 25%) was effective as of March 4, before it was increased to 35% on August 1.) 10% on energy imports effective as of March 4, 2025; USMCA exception and 10% on potash imports effective as of March 7, 2025.



Thank You!